System - a combination of one or more objects. The objects that make up a system do not have an internal structure.

Closed System

- No outside (external) forces act on the system
- Energy and momentum are conserved

Open System

- There are outside (external) forces act on the system
- Energy and momentum are NOT conserved
    - *Energy changes due to work done by the external force:*

      $\Delta E = W_{ext}$

      $\Delta E = F \cdot \Delta x$
    - *Momentum changes due to the impulse imparted by the external force:*

      $\Delta p = impulse$

      $\Delta p = F \cdot \Delta t$

Whether a system is open or closed depends on how the system is defined.

Single Object System
If the system consists only of a single object, any force acting on it is external and will change the energy and momentum of the system.

Object-Earth System
If the system consists of an object and the Earth, then any force other than gravity acting on it is external and will change the energy and momentum of the system.

Multi-Object System
If the system consists of multiple objects (as in collisions and explosions, for example), any force other than the interaction of the objects with each other is external and will change the energy and momentum of the system.

Multi-Object and String System
If the system consists of object connected by a string, any force other than the tension in the string is external and will change the energy and momentum of the system.

Friction
Unless the ***surface*** on which the object is sliding is specifically included in the system, friction between the object and the surface is an external force and will decrease the energy and momentum of the system.

[***Note***: including the Earth in the system does *not* count as including the surface on which the object is moving in the system]